

FYKE SIMON V. TONEL

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Education

STI College Caloocan

2022 – Present

Bachelor of Science in Computer Science

Caloocan City, Metro Manila

- Current Cumulative GWA: 1.27
- Best Thesis, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024-25
- Best Methods of Research Project, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024–25
- Entrant, Codefest (Local – Tagisan ng Talino), 2024–25
- President’s Honor (STI Academic Center – Academic Honors), 2022 – Present
- Alumni Scholar (Awarded for High Honors in Senior High; 50% scholarship in college)

STI College Caloocan

2020 – 2022

Senior High School (Information & Communications Technology)

Caloocan City, Metro Manila

- Graduated with High Honors
- Completion of Computer Programming – Java and .NET Technology (IT in Mobile App & Web Development)

Lakan Dula High School

2016 – 2020

Junior High School

Manila City, Metro Manila

- Graduated with Honors

Projects

Flick Player: Music Player with Custom UAC 2.0

January 2026

Flutter, Dart, Rust, Kotlin, DSP

Lead Developer – Maintainer

- Engineered a high-performance audio engine in Rust with a custom USB Audio Class (UAC) 2.0 implementation, enabling **bit-perfect playback through external DACs/AMPs** by bypassing Android’s AudioFlinger.
- Built a dual-engine system (Rust + just_audio) with automatic fallback to ensure compatibility across devices with and without USB audio support.
- Designed a real-time audio processing pipeline supporting **EQ, dynamics, spatial effects, gapless playback, and crossfade** with minimal latency.
- Implemented a Flutter ↔ Rust bridge via FFI for **thread-safe, low-latency communication** between UI and native audio engine.
- Optimized large-scale music library handling with fast scanning, metadata extraction, and indexed queries for smooth performance.
- Developed an adaptive, high-refresh-rate UI with waveform visualization and immersive playback controls tailored for audiophiles.
- **Released as a free and open-source project with an active user base with 560+ downloads and early community adoption.**

br41ndmg: High-Fidelity Audio Resampling Library

Rust (DSP, Audio Processing)

Lead Developer

- Developed a high-performance audio resampling library using **polyphase sinc interpolation** for accurate sample-rate conversion.
- Implemented configurable FIR filters with multiple window functions (**Hann, Hamming, Blackman, Kaiser**) for precise frequency response control.
- Supported **arbitrary sample rate conversion** (e.g., 44.1kHz → 48kHz) with low aliasing and high stopband attenuation.

- Designed both **offline and real-time streaming resampling pipelines** for audio applications.
- Validated DSP accuracy through **impulse, sine, and sweep testing**, with benchmarked performance using Criterion.
- Achieved high-quality targets including **>100 dB stopband attenuation and <0.1 dB passband ripple**.

PASADA: Ride-Hailing & Fleet Management Ecosystem

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, Google Cloud Platform

Lead Full-Stack Developer

- Architected a modernized public transport solution interconnecting four distinct applications: Pasada Passenger, Pasada Driver, Pasada Admin, and the Pasada Official Website.
- Engineered a custom analytics pipeline for the Admin Dashboard that provides 7-day predictive forecasts for both cooperative booking frequency and the route traffic.
- Integrated the Gemini API within both Passenger and Admin applications to interpret complex, in-house predictive and real-time traffic data into easily digestible, actionable, summaries for users and fleet managers.
- Developed the Passenger and maintained the Driver mobile applications using Flutter, featuring real-time, Google Maps-sourced traffic analytics for optimized routing and planning.
- Built the official website using Next.js and Tailwind CSS, featuring a live public ride-tracker implemented with MapLibre GL JS and OpenStreetMaps.

Mochi: Family-Shared AI Companion with Persistent Memory System

April 2026

Flutter, Node.js, SQLite, llama.cpp, Gemini API, Termux, Kotlin

Lead Developer

- Designed a shared AI system where a single evolving pet maintains memory, mood, and relationship states across multiple users.
- Architected a hybrid AI pipeline using local inference (TinyLlama via llama.cpp) with cloud fallback (Gemini API).
- Built a backend managing conversation history, memory injection, mood tracking, and affection scoring.
- Implemented real-time synchronization via WebSockets across client devices.
- Deployed using a low-cost home server setup (Termux + Cloudflare Tunnel) for always-on availability.

Locker: Advanced Media Hiding & Security Application

November 2025

Flutter, Dart, Kotlin

Lead Developer – Maintainer

- Developed a secure media vault application for Android with **biometric authentication, PIN/password protection, and optional AES-256 encryption**.
- Implemented a **decoy vault system and auto-kill feature** to protect user privacy under forced access scenarios.
- Built a full media pipeline including import, compression, secure storage, and recovery with support for images, videos, documents, and archives.
- Designed efficient file handling with **duplicate detection, tagging, search, and album organization** for large datasets.
- Engineered performance optimizations (frame rate control, compression strategies, memory management) to maintain smooth UX under heavy workloads.
- **Published as a free and open-source application with real users and ongoing usage.**

Revo: Productivity & Study Companion Application

December 2025

React, Next.js, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Spotify Web API

Lead Developer

- Developed a productivity application featuring an integrated calendar, note-taking system, and study-focused timer.
- Implemented multiple learning techniques including Pomodoro, 52/17, Deep Work, Short Burs, and custom study cycle configuration.
- Integrated Spotify Web API to allow users to sync and stream their personal music libraries within the app.

Experience

Back-End Developer (Intern)

January 2026 – February 2026

Inspire Holdings Inc.

- Serves as the Technical Lead for *Inspire Wallet* and *Admin 2.0*, a FinTech investment platform, overseeing the full backend lifecycle and architectural decisions.
- Architected a scalable backend infrastructure using NestJS and AWS, implementing load balancing, rate limiting, and local/remote caching via Redis to ensure high availability.
- Engineered the payment gateway integration for versions 1.0 and 2.0 using the UnionBank API, ensuring secure and seamless financial transactions.
- Optimized database performance and management using Supabase and ORMs, while handling transactional emails via Resend.
- Wrote comprehensive unit tests to validate core logic and security features, and collaborated cross-functionally to ensure UI/UX quality standards were met.
- Defined and implemented the company's first proper SDLC and a collaborative QA testing protocol involving cross-team validation to reduce bugs in production.

Software Developer (Intern)

February 2026 – March 2026

Socia I.T. Solutions

Projects

LootBX Mobile: Real-Time Streaming & Interactive Rewards Platform

February 2026

React Native, Expo, TypeScript, JavaScript, MongoDB

Full-Stack Developer

- Contributed to the development of high-performance API integrations for the **Lootbx Mobile** ecosystem, ensuring seamless data flow and an enhanced user experience.
- Engineered complex logical state management for core functions and components during the architectural transition from web-based platforms to a unified mobile environment.
- Collaborated cross-functionally within a high-velocity startup environment to implement the full-stack Lootbx streaming platform, prioritizing scalability and low-latency performance.
- Optimized the SDLC and architectural decisions for the broader Lootbx ecosystem, leveraging TypeScript and modern backend frameworks to ensure robust production standards.

Papa Burger Ecosystem: Multi-Platform Restaurant Operations System

February 2026 – March 2026

React TSX, PHP, PostgreSQL, Next.js, Flutter, Dart

Full-Stack Developer

- Built the entire front-end architecture for a large-scale restaurant ecosystem, integrating multiple backend services.
- Developed an enterprise-level platform including POS system, Driver Portal, Landing Page, and Franchising modules.
- Designed and implemented scalable UI systems with modular components and maintainable state management.
- Integrated and tested APIs to ensure reliable transactions, order flows, and operational efficiency across modules.

Skills

Core Languages

Dart, Rust, Kotlin, TypeScript, Java, SQL

Systems & Backend Engineering

Node.js (NestJS, ExpressJS), Bun, Deno

PostgreSQL, SQLite, Supabase, Redis, QuestDB

WebSockets, API design, RBAC, rate limiting

Mobile & Frontend Development

Flutter, React, Next.js, Remix

Tailwind CSS, MapLibre GL JS

Low-level & Performance

Rust (systems programming, DSP, audio processing)

FFI (Flutter ↔ Rust), real-time pipelines, memory optimization

Cloud & Infrastructure

Docker, Vercel, Fly.io, Cloudflare, GCP

Event-driven systems (SQS/SNS), caching, deployment pipelines

AI & Applied ML

llama.cpp, Gemini API

Tools & Environments

Linux (Arch Linux and CachyOS), Neovim, Git, GitHub, GitLab, Postman

Android Studio, Neovim, Zed

Figma

Certifications

Learn Dart: First Step to Flutter

July 2024

Educative.io

Java Fundamentals

June 2023

Oracle